



Spontaneous regression rate of high-grade cervical lesions in women under 35 at Dijon University Hospital: Incidence and predictive factors

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ABSTRACT

Introduction: According to current guidelines, women with CIN 2/3 should undergo a conization procedure. However, for young patients with small lesions, it is possible to consider therapeutic abstention with close monitoring while awaiting potential spontaneous regression. The objectives of this study were to evaluate the regression rate of CIN 2/3 lesions over 24 months of follow-up in women under the age of 35 and to identify predictive factors for spontaneous healing.

Methods: This was a retrospective observational study based on data from patients diagnosed with CIN 2 or CIN 3 who underwent close monitoring every 6 months via colposcopy, for a maximum follow-up period of 24 months between May 2008 and May 2024. The primary endpoint was the rate of spontaneous regression of high-grade lesions.

Results: 104 patients, with a mean age of 27.8 years, were included. Cervical intraepithelial neoplasia spontaneously regressed in 37 patients (35.6 %, 95 % CI: 26.4 % – 45.5 %) after a median follow-up of 25.22 months. In multivariate analysis, patients with a lesion limited to a single quadrant were more likely to experience spontaneous healing (OR = 2.98 [1.21–7.34], $p = 0.017$). No other presumed predictive factors were found to significantly influence spontaneous regression. 2 patients had an in-situ cancer at conization, performed after surveillance and no cases of invasive cancer were described.

Conclusion: The results of the study are consistent with the literature and support the possibility of 24-month surveillance of patients with a high-grade lesion, especially given the low risk of progression to more severe disease within two years.

Introduction

Persistent infection with high-risk types of human papillomavirus (HPV), particularly types 16 and 18, is the principal risk factor for the development of cervical cancer. In the majority of cases, HPV infection is cleared by the host immune response without clinical consequences. However, approximately 10 % of infections persist, increasing the risk of progression to precancerous cervical intraepithelial neoplasia (CIN) [1].

The classical model of cervical carcinogenesis describes a sequential progression from normal squamous epithelium to CIN 1, CIN 2, and finally CIN 3 [2]. A more recent classification distinguishes between two groups of lesions based on their biological behavior and risk of progression: low-grade intraepithelial lesions (CIN 1) and high-grade intraepithelial lesions (CIN 2 and CIN 3) [3].

Current clinical guidelines in France recommend the treatment of high-grade lesions by conization [4]. However, this surgical procedure is

associated with non-negligible obstetric risks, particularly an increased risk of preterm delivery, as demonstrated by Kyrgiou and Arbyn [5]. Hence, conservative management options are increasingly being considered, especially for women who have not completed their reproductive plans and in whom the risk of progression is deemed low.

Several studies have reported spontaneous regression rates of 43–63 % for CIN 2 and 28–36 % for CIN 3 over two years [6–14]. In light of this, the 2016 guidelines from the French National Cancer Institute (INCa) allow for therapeutic abstention in selected women under 30 years old, provided the lesion is limited in extent, without signs of invasion, and with a fully visible squamocolumnar junction. This approach is also endorsed by international guidelines, including those from the American College of Obstetricians and Gynecologists (ACOG) and the British Society for Colposcopy and Cervical Pathology (BSCCP), which recommend close surveillance for up to 24 months in appropriate cases [10–12].

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Objectives

The primary objective of this study is to evaluate the rate of spontaneous regression of histologically confirmed high-grade cervical lesions (CIN 2 and CIN 3) at 24 months in women under 35 years old.

Materials and methods

Study design

This was a monocentric, retrospective, observational, and descriptive study conducted at the University Hospital of Dijon, a regional reference center for colposcopy, over a 16-year period from May 2008 to May 2024.

Study population

The study population consisted of asymptomatic women under the age of 35 diagnosed with high-grade cervical intraepithelial neoplasia (CIN 2 or CIN 3) identified via colposcopy, following abnormal cervical cytology (FCV) and/or high-risk human papillomavirus test (HR-HPV) performed as part of the French organised cervical cancer screening programme (cytological examination between the ages of 25 and 29 and HR-HPV testing between the ages of 30 and 35). Only patients who opted for close colposcopic surveillance were included.

Exclusion criteria were:

- A type 3 transformation zone on colposcopy;
- Clinical or histopathological severity criteria necessitating immediate ablative treatment;
- Prior conization procedures.

At baseline, all patients underwent colposcopy with biopsy. Follow-up included colposcopy with biopsy every six months for two years. At the end of the surveillance period, ablative treatment was performed in cases of persistent high-grade lesions or earlier in cases of progression, such as the appearance of adenocarcinoma in situ or invasive carcinoma.

Data collection

Data were collected retrospectively from electronic medical records and, when necessary, supplemented by telephone interviews. The variables recorded included:

- Demographic data: age, HPV vaccination status, smoking status;
- Colposcopic data: number of quadrants involved, visibility of the transformation zone;
- Histopathological data: results of cytology and biopsies;
- Virological data: HPV genotype.

Colposcopy and histological assessment

All colposcopies were performed by two experienced colposcopists, Professor S. Douvier and Dr. C. Bouit, both members of the French Society of Colposcopy and Cervico-Vaginal Pathology (SFCPCV). Biopsy specimens were evaluated by either the Pathology Department of the University Hospital of Dijon or the private pathology laboratory CYPATH Dijon.

To improve diagnostic accuracy and reproducibility in grading CIN lesions, all biopsy samples were subjected to immunohistochemical analysis using p16 and Ki67 markers.

Primary objective

The primary objective of this study was to evaluate the spontaneous cure rate of high-grade cervical intraepithelial neoplasia (CIN 2 or CIN

3) in women under 35 years of age who were managed conservatively through close colposcopic surveillance.

The primary endpoint was the spontaneous cure, defined as either:

- Disappearance of the high-grade lesion: complete resolution of colposcopic abnormalities with confirmation of normal histology; or
- Regression of the lesion: reduction in severity of the colposcopic findings associated with histological confirmation of a low-grade lesion (CIN 1).

Additionally, if a patient underwent ablative treatment during or at the end of follow-up, and histological analysis of the excisional specimen revealed a low-grade lesion or less, the case was retrospectively classified as a spontaneous cure.

Secondary objectives

The secondary objective was to identify predictive factors associated with spontaneous healing of CIN 2/CIN 3 lesions. The following variables were analyzed:

- HPV vaccination status
- Smoking status
- Age at diagnosis
- Immunosuppression (e.g., due to HIV infection or immunosuppressive therapy)
- Presence of HPV 16 or 18
- Extent of cervical involvement (number of quadrants affected)
- Type of transformation zone (Type 1 or 2)
- Pregnancy during follow-up
- Cytological-histological dissociation (discrepancy between cytology and biopsy results)

Statistics

Quantitative and qualitative variables were categorized based on established literature or clinical relevance. Continuous variables were described using means and standard deviations for normally distributed data, and medians with interquartile ranges (IQRs) for non-normally distributed data. Categorical variables were summarized as frequencies and percentages.

Comparisons of categorical variables were performed using the Chi-square test or Fisher's exact test, as appropriate. Comparisons of continuous variables were conducted using parametric or non-parametric analysis of variance, depending on the distribution and variance homogeneity.

Time-to-event analyses (e.g., time to spontaneous regression or cure) were assessed using Kaplan–Meier survival curves. Multivariable analyses were performed using logistic regression for binary outcomes and Cox proportional hazards models for time-to-event outcomes. A robust variance estimator was applied to correct for potential heteroskedasticity.

Non-linear relationships between continuous covariates and the outcome were examined using fractional polynomials. The discrimination ability of the logistic regression model was evaluated using the area under the receiver operating characteristic curve (AUC), and calibration was assessed with the Hosmer–Lemeshow goodness-of-fit test.

For the Cox model, assumptions of log-linearity and proportional hazards were tested. Log-linearity was assessed via fractional polynomials, and the proportional hazards assumption was verified using Schoenfeld residuals.

All statistical analyses were performed using Stata software, version 18 (StataCorp, College Station, TX, USA). A p-value < 0.05 was considered statistically significant.

Results

Population characteristics

Between May 2008 and May 2024, a total of 128 patients were enrolled in colposcopic surveillance for histologically confirmed high-grade cervical intraepithelial neoplasia (CIN 2 or CIN 3). Of these, 24 patients were excluded due to incomplete follow-up, leaving 104 patients eligible for analysis. Demographic, clinical, and histopathological characteristics are summarized in Table 1.

The mean age at the time of the initial colposcopy was 27.8 years (SD = 3). No patient was in the cohort immunosuppressed. Nineteen patients were vaccinated against HPV, and the vaccination status was unknown in 39 patients. 52 patients were smokers at the start of their treatment, and of these, 15 stopped smoking during follow-up. HPV vaccination had been administered in 19 patients (18.3 %), while vaccination status was unknown in 39 cases (37.5 %). Fifty-two patients (50 %) were active smokers at baseline, of whom 15 (28.8 %) ceased smoking during the surveillance period. Cytological screening revealed LSIL in 34.62 % of cases, HSIL in 25 %, ASC-US in 22.12 %, ASC-H in 13.46 % and AGC in 0.96 %. Eighty patients had CIN 2 and twenty-four patients had CIN 3. Most often, patients had a lesion occupying one quadrant (41.35 %) or two quadrants (42.31 %). In 54.81 % of cases, the transformation zone was type 1, and in 41.35 % type 2. HPV 16 and 18 were identified in 33 patients. Twenty-five patients achieved full-term pregnancy during surveillance. Initial histological classification revealed CIN 2 in 80

Table 1
Description of study population.

	Total population	Spontaneous healing	Non spontaneous healing
N (%)	104 (100 %)	37 (35,6 %)	67 (46,4 %)
Age of 1 st colposcopy (years)	27,8	27,2	28,2
Vaccinal status			
Vaccinated	19 (18,27 %)	10 (27,03 %)	9 (13,43 %)
Unvaccinated	46 (44,23 %)	18 (48,65 %)	28 (41,79 %)
Missing data	39 (37,5 %)	9 (24,32 %)	30 (44,78 %)
Smoking status			
Smoking	52 (50 %)	16 (43,24 %)	36 (53,73 %)
Non-smoking	49 (47,12 %)	20 (54,05 %)	29 (43,28 %)
Missing data	3 (2,88 %)	1 (2,7 %)	2 (3 %)
Smoking cessation	15/52 (28,85 %)	6/16 (37,5 %)	9/36 (25 %)
Cytology			
ASC-US	23 (22,12 %)	12 (32,43 %)	11 (16,42 %)
ASC-H	14 (13,46 %)	4 (10,81 %)	10 (14,93 %)
AGC	1 (0,96 %)	1 (2,70 %)	0 (0 %)
LSIL	36 (34,62 %)	11 (29,73 %)	25 (37,31 %)
HSIL	26 (25 %)	7 (18,92 %)	19 (28,36 %)
Missing data	4 (3,85 %)	2 (5,41 %)	2 (2,99 %)
Cervical intraepithelial neoplasia			
CIN 2	80 (76,92 %)	31 (83,78 %)	49 (73,13 %)
CIN 3	24 (23,08 %)	6 (16,22 %)	18 (26,87 %)
Quadrant			
1	43 (41,35 %)	22 (59,46 %)	21 (31,34 %)
2	44 (42,31 %)	11 (29,73 %)	33 (49,25 %)
3	8 (7,69 %)	2 (5,41 %)	6 (8,96 %)
4	8 (7,69 %)	1 (2,70 %)	7 (10,45 %)
Missing data	1 (0,96 %)	1 (2,70 %)	0 (0 %)
Transformation zone			
ZT1	57 (54,81 %)	24 (64,86 %)	33 (49,25 %)
ZT2	45 (41,35 %)	12 (32,43 %)	33 (49,25 %)
Missing data	2 (1,92 %)	1 (2,7 %)	1 (1,49 %)
HPV genotype			
HPV 16, 18	33 (31,73 %)	5 (7,46 %)	28 (41,79 %)
Missing data	42 (40,38 %)	19 (51,35 %)	13 (19,40 %)
Cytohistological discordance	6 (5,77 %)	0 (0 %)	6 (8,96 %)
Pregnant	25 (24,04 %)	4 (10,81 %)	21 (31,34 %)
Conization	54 (51,92)	2 (5,41)	52 (77,61)
Lost from sight	15 (14,42 %)	0 (0 %)	15 (22,39)

patients (76.9 %) and CIN 3 in 24 patients (23.1 %). Lesions most commonly involved one (41.4 %) or two (42.3 %) cervical quadrants. The transformation zone was classified as type 1 in 54.8 % of cases and type 2 in 41.4 %. High-risk HPV genotypes 16 and/or 18 were identified in 33 patients (31.7 %). Notably, 25 patients (24 %) became pregnant and delivered at term during the surveillance period.

Among the 104 patients under surveillance, 37 patients (35.6 %) demonstrated spontaneous healing, defined as lesion disappearance or regression—yielding a healing rate of 35.6 % (95 % confidence interval [CI]: 26.4 %–45.5 %). i.e. a healing rate of 35.6 % (95 % CI: 26.4 % - 45.5 %). Thirty-one patients with CIN 2 lesions achieved spontaneous healing (38.75 % of CIN 2 lesions (95 % CI: 28.06 % - 50.30 %), in CIN3 subgroup 6 of 24 patients (25.0 %; 95 % CI: 9.8 %–46.7 %) experienced spontaneous healing as did 6 patients with CIN 3 lesions, representing a cure rate of 25 % of CIN 3 lesions (25 %; 95 % CI: 9.77 % - 46.71 %).

A total of 54 patients (51.9 %) required ablative conization during follow-up due to lesion persistence or progression. Fifty-four patients underwent ablative conization treatment during surveillance. The final anatomopathological results of these patients, described in Table 2, were CIN 2 in 35.19 % and CIN 3 in 57.45 %. In-situ adenocarcinoma (3.7 %) was identified in 2 patients. No cases of invasive cancer were detected.

Final histopathological examination of the excisional specimens is presented in Table 2: CIN 2: 35.2 %; CIN 3: 57.4 %; Adenocarcinoma in situ: 3.7 % (n = 2); No cases of invasive carcinoma were identified.

Univariate and multivariate analyses

No statistically significant associations were observed between spontaneous healing and patient age, HPV vaccination status, smoking status, or transformation zone type in either univariate or multivariate analyses (Tables 3 and 4).

In contrast, involvement of a single cervical quadrant was significantly associated with a higher likelihood of spontaneous healing. Patients with lesions confined to one quadrant were nearly 3 times more likely to experience spontaneous healing compared to those with lesions affecting two or more quadrants (adjusted odds ratio [aOR] = 2.98; 95 % CI: 1.21 – 7.34; p = 0.017).

Pregnancy during surveillance was associated with a lower rate of spontaneous healing in univariate analysis; however, this association did not remain statistically significant after adjustment in multivariate models.

Analyses examining the influence of HPV genotypes 16 and 18, as well as cytological-histological dissociation, were not performed due to insufficient sample sizes and missing data.

Time-dependent multivariate analyses and survival curve

The mean duration of follow-up in our cohort was 25.22 months. The overall time-to-event analysis, from initial diagnosis to spontaneous healing or end of follow-up, is illustrated in Fig. 1.

Among patients who experienced spontaneous healing, 50 % had achieved healing by 6 months of follow-up, as shown in Fig. 2.

Table 2
Definitive anatomopathological results after conization performed during or at the end of surveillance in the absence of spontaneous healing.

	Conization
N (%)	54 (51,92 %)
Normal/Cervicitis	1 (1,85 %)
CIN 1/Low grade	1 (1,85 %)
CIN 2	19 (35,19 %)
CIN 3	31 (57,45 %)
AIS	2 (3,7 %)
Invasive adenocarcinoma	0 (0 %)
Invasive carcinoma	0 (0 %)

Table 3
Univariate analyses of spontaneous healing of CIN 2 and CIN 3 lesions in the study population.

Criteria	No spontaneous healing	Spontaneous healing	p
N (%)	67 (64,4 %)	37 (35,6 %)	
Age of 1 st colposcopy (standard deviation)	28,2 years (3,3)	27,2 years (2,3)	0,08*
Vaccinated	9 (13,43 %)	10 (27,03 %)	0,411†
Smoking	36 (53,73 %)	16 (43,24 %)	0,308†
Smoking cessation	9/36 (25 %)	6/16 (37,5 %)	0,502†
Quadrant			
1	21 (31,34 %)	22 (59,46 %)	0,006††
2	33 (49,25 %)	11 (29,73 %)	0,094†
3	6 (8,96 %)	2 (5,41 %)	0,710†
4	7 (10,45)	1 (2,70 %)	0,256†
Transformation zone			
ZT1	33 (49,25 %)	24 (64,86 %)	0,144†
ZT2	33 (49,25 %)	12 (32,43 %)	0,212†
Pregnancy	21 (31,34 %)	4 (10,81 %)	0,030††

*Chi2 test
†Fisher exact test
‡Significant difference p < 0,05

Table 4
Multivariate analyses of spontaneous healing of CIN 2 and CIN 3 lesions in the study population, logistic regression model.

Criteria	Odds Ratio [IC 95 %]	p
Quadrant 1	2,98 [1,21–7,34]	0,017*†
Pregnancy	0,22 [0,08–1,16]	0,029*

*Chi2 test
†Significant difference p < 0,05

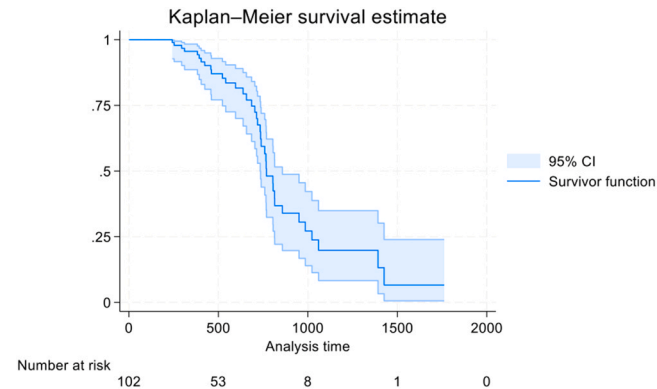


Fig. 1. Survival curve evaluating the time from initial diagnosis of a high-grade lesion to the end of surveillance for the study population.

In multivariate Cox proportional hazards analyses, patient age, smoking status, HPV vaccination status, transformation zone type, and pregnancy during surveillance were not significantly associated with the time to spontaneous cure.

Conversely, lesions confined to a single cervical quadrant demonstrated a significantly shorter time to spontaneous healing compared to lesions involving two or more quadrants (Table 5), indicating faster regression in less extensive disease.

Discussion

In our study, spontaneous healing was observed in 35.6 % of high-grade cervical intraepithelial neoplasia (CIN 2 and CIN 3) cases, with rates of 38.75 % in CIN 2 lesions and 25 % in CIN 3 lesions. The surveillance period was extended to 24 months, in accordance with

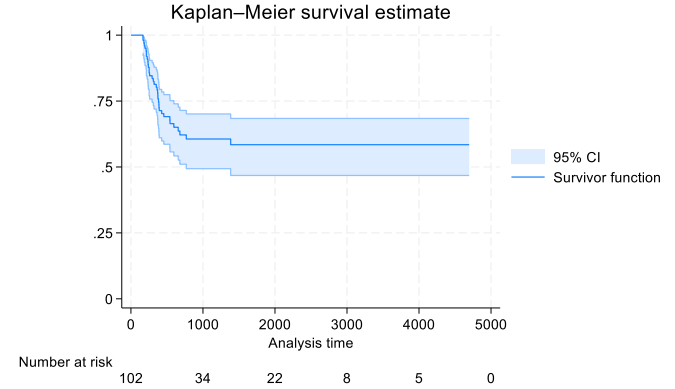


Fig. 2. Survival curve evaluating the time between the initial diagnosis of a high-grade lesion and spontaneous healing during surveillance for the 37 patients with spontaneous healing.

Table 5
Multivariate analysis of spontaneous healing of CIN 2 and CIN 3 lesions in the study population, logistic regression model as a function of time.

Criteria	Hazard Ratio [IC 95 %]	p
Quadrant 1	3,42 [1,53–7,62]	0,003*†

*Cox model
†Significant difference p < 0,05

recommendations from the *Institut National du Cancer* (INCa) [4]. Among patients who experienced spontaneous healing, half had achieved this by 6 months' follow-up.

Among patients who did not undergo spontaneous healing and subsequently received conization, only two patients (1.9 % of the total cohort) were found to have adenocarcinoma in situ, and no invasive cancers were detected. Importantly, lesions confined to a single quadrant at colposcopy were significantly more likely to heal spontaneously than lesions involving multiple quadrants, whereas other studied factors showed no significant association.

Our findings align with previous literature [6–9], supporting a policy of 24-month surveillance for patients under 30 years with high-grade cervical lesions. The main rationale is to avoid systematic conization and its potential obstetric complications [5], particularly relevant in France, where the average age of first pregnancy is decreasing (estimated at 31 years according to INSEE 2023 data [15]). Patients can still conceive during the surveillance period, allowing pregnancy without the risks associated with premature treatment.

While high-grade lesions confer a risk of progression to invasive cancer, the incidence of severe progression during surveillance remains low. In our cohort, few patients progressed to more severe lesions, consistent with a 0.3 % progression rate reported in the literature [9]. Long-term retrospective data from New Zealand show that approximately 31 % of untreated CIN 3 lesions progress to invasive cancer over 30 years, with the risk increasing to 50 % if cytology remains positive within 6–24 months post-biopsy [16,17]. This underscores the importance of timely ablative treatment beyond 24 months if lesions persist, as supported by the meta-analysis of Tainio et al., which showed increased progression rates after 2 years of persistent disease. Especially since their subsequent regression rate after 2 years is low [7]. In the event of AIS, ablation treatment must be performed [4].

Regarding predictors of spontaneous healing, only the extent of cervical involvement (number of quadrants affected) was significant in our study. Other studies have suggested that persistent HPV infection—particularly HPV 16 and 18—and HSIL cytology are associated with lower regression rates [18,19]. Age appears to have only a modest and inconsistent influence, insufficient to dictate specific age thresholds for treatment [6,7]. Thus, surveillance could reasonably be extended to

patients over 35 who have not completed childbearing. Since reliable predictive factors remain elusive, some authors propose algorithms incorporating clinical and virological factors to help identify patients who might benefit from early conization [19].

Our study's strengths include its monocentric, intention-to-treat design, with all colposcopies performed by two experienced colposcopists and consistent use of immunohistochemical diagnostic methods for CIN. The extended 24-month follow-up period offers more comprehensive insight into spontaneous healing rates compared to many prior studies [10–14].

However, the study has notable limitations. The retrospective design contributed to missing data, especially regarding vaccination status and HPV genotyping, despite efforts to retrieve information by patient contact. Additionally, a 14.4 % loss to follow-up rate highlights challenges in patient compliance with prolonged surveillance—a critical consideration for this management approach. Surveillance is only effective if patients understand and adhere to the protocol, which must be emphasized by clinicians.

Future research should focus on identifying new predictive factors for spontaneous healing, particularly the role of HPV vaccination as coverage increases and impacts HPV-related disease prevalence. Moreover, long-term outcomes for patients who experience spontaneous regression remain underexplored, especially when compared with those treated by conization, who face a higher risk of recurrence [20,21].

Conclusion

Our results corroborate existing evidence supporting a 24-month surveillance strategy for young patients with high-grade cervical lesions, given the low risk of progression to invasive disease within this timeframe. We recommend offering colposcopic surveillance every 6 months for up to 24 months to women under 30 years of age with high-grade lesions who are planning pregnancy, in the absence of severity criteria and provided that patients consent to this conservative management approach.

CRediT authorship contribution statement

Bouit Camille: Supervision, Methodology. **Tronche Perrine:** Writing – review & editing, Resources, Methodology, Data curation. **Douvier Serge:** Validation. **Aho Serge:** Formal analysis. **Kadhel Philippe:** Validation.

Declaration of Competing Interest

The authors declare that they have no known financial conflicts of interest or personal relationships that could have influenced the work presented in this article. The authors are not members of the editorial board of this journal and did not participate in the editorial review or decision to publish this article. The authors declare the following financial interests/personal relationships that could be considered potential conflicts of interest: non-financial support for statistical analysis and editorial assistance.

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